

Personalised therapy and target report

Screening report

Retinal screening to therapy options, target priorities and trials

Patient	64 years
Current medicines	pioglitazone, semaglutide
Model	insightrx-dinov2L-lora-ens2-141f4d6893 (live)
Status	Unsaved screening (held 1 hour). Save the patient to consult and track.

Summary

- Retina:** Referable DR signal.
- Findings:** Referable diabetic retinopathy, Macular edema signal, Blood pressure and vascular risk.
- Therapy options:** 6 guideline classes; **2** major alert(s) on current medicines.
- Top target:** VEGFA (priority 98): Referable diabetic retinopathy (validated retinal model); Anti-VEGF intravitreal therapy: specialist-administered; 6 approved drugs; drug-contact site pLDDT 95 (PDB 1CZ8).
- Trials:** 9 recruiting trial(s) matched by finding.
- CMS diabetic eye exam gap:** open. Screening captured; the gap closes once an eye care professional reads it.

Retina

Eye	Result	Referable DR score	Highest ICDR grade	Macular edema signal
Right eye	Referable DR signal	1.00	2	yes
Left eye	Referable DR signal	1.00	2	yes



Right eye, quality assessable, DR score 1.00



Left eye, quality assessable, DR score 1.00

Whole-body signals

Research association models from the retina plus patient details; history recorded by the clinician takes precedence. Red bar: score at or above the frozen threshold (black tick).

Condition	Output	Score vs threshold	Reliability
Hypertension	Research signal		Passes the release gate: cross-validated AUROC 0.67 (95% CI 0.64 to 0.70)
Diabetic kidney involvement	No exploratory signal		Exploratory: cross-validated AUROC 0.60 (95% CI 0.51 to 0.68), below the release gate
Diabetes-related vascular disease	No exploratory signal		Exploratory: cross-validated AUROC 0.59 (95% CI 0.54 to 0.63), below the release gate
Prior myocardial infarction	No exploratory signal		Exploratory: cross-validated AUROC 0.62 (95% CI 0.56 to 0.67), below the release gate

Condition	Output	Score vs threshold	Reliability
Diabetic neuropathy	No exploratory signal		Exploratory: cross-validated AUROC 0.59 (95% CI 0.52 to 0.66), below the release gate
Diabetic foot	No exploratory signal		Exploratory: cross-validated AUROC 0.61 (95% CI 0.57 to 0.66), below the release gate
Obesity	Near-chance score		Near chance: cross-validated AUROC 0.53 (95% CI 0.47 to 0.58 includes 0.5)
Cardiovascular composite (vascular disease or prior MI)	No exploratory signal		Exploratory: cross-validated AUROC 0.59 (95% CI 0.55 to 0.63), below the release gate
Diabetes complications composite (kidney, nerve or foot)	No exploratory signal		Exploratory: cross-validated AUROC 0.55 (95% CI 0.51 to 0.59), below the release gate

Therapy options to consider

Guideline classes for this patient's findings, ranked by guideline strength only (sponsorship never enters the ranking). Current medicines: pioglitazone, semaglutide.

Therapy class	Why (findings)	Evidence tier	Example drugs	Already on	Source
Tighten glucose, blood pressure and lipids	Referable diabetic retinopathy	Guideline first-line	-	-	ADA Standards of Care in Diabetes, Section 12 (Retinopathy)
Anti-VEGF intravitreal therapy	Referable diabetic retinopathy, Macular edema signal	Specialist-administered	aflibercept, ranibizumab, faricimab, bevacizumab	-	AAO Diabetic Retinopathy Preferred Practice Pattern; DRCR Retina Network Protocols T and S
Fenofibrate (adjunct)	Referable diabetic retinopathy	Trial evidence	fenofibrate	-	FIELD (Lancet 2007); ACCORD-Eye (NEJM 2010); LENS (NEJM Evidence 2024)
ACE inhibitor or ARB	Blood pressure and vascular risk	Guideline first-line	lisinopril, ramipril, losartan, valsartan, candesartan	-	ADA Standards of Care, Sections 10 and 11; KDIGO 2022 Diabetes in CKD
SGLT2 inhibitor	Blood pressure and vascular risk	Guideline first-line	empagliflozin, dapagliflozin, canagliflozin	-	KDIGO 2022 Diabetes in CKD; ADA Standards of Care, Section 11
GLP-1 receptor agonist	Blood pressure and vascular risk, Diabetes control	Guideline first-line	semaglutide, dulaglutide, liraglutide	semaglutide	ADA Standards of Care, Sections 9 and 10; SUSTAIN-6; FLOW (NEJM 2024)

Interaction and drug-disease alerts

Severity	Alert	Involves	Detail and source
Major	Diabetic retinopathy complications	semaglutide + dulaglutide + liraglutide (already prescribed)	Semaglutide was associated with more DR complications in SUSTAIN-6, linked to rapid glucose lowering. With a retinopathy signal, arrange retinal follow-up before and after starting. Semaglutide label, Warnings: diabetic retinopathy complications; ADA Standards of Care, Section 12
Major	Macular edema with pioglitazone	pioglitazone (already prescribed)	Thiazolidinediones have been associated with new or worsening macular edema. With a macular edema signal, review the need for pioglitazone and refer. Pioglitazone label, Warnings: macular edema

Personalised target priorities

Where this patient's phenotype points at the molecular level. Priority = 100 x phenotype link x (0.5 guideline actionability + 0.3 structural tractability at the drug site + 0.2 clinical fit). Actionability comes from US guideline strength, so drugs approved only abroad do not lift a target. It ranks where to look; it is not a treatment recommendation.

Top target for this patient: VEGFA (Vascular endothelial growth factor A), priority 98. Referable diabetic retinopathy (validated retinal model); Anti-VEGF intravitreal therapy: specialist-administered; 6 approved drugs; drug-contact site pLDDT 95 (PDB 1CZ8).

Target	Priority	Linked findings	Drugs	Structure	Rationale
VEGFA Vascular endothelial growth factor A	 98	Referable diabetic retinopathy; Macular edema signal	6 approved 3 pipeline	drug-contact site pLDDT 95 (PDB 1CZ8)	Classes: Anti-VEGF intravitreal therapy
AGTR1 Angiotensin II receptor type 1	 84	Blood pressure and vascular risk	7 approved 0 pipeline	drug-contact site pLDDT 95 (PDB 4ZUD)	Classes: ACE inhibitor or ARB
ACE Angiotensin-converting enzyme	 83	Blood pressure and vascular risk	10 approved 0 pipeline	drug-contact site pLDDT 93 (PDB 1O86)	Classes: ACE inhibitor or ARB
SLC5A2 Sodium/glucose cotransporter 2 (SGLT2)	 83	Blood pressure and vascular risk	8 approved 0 pipeline	drug-contact site pLDDT 92 (PDB 7VSI)	Classes: SGLT2 inhibitor
ANGPT2 Angiopoietin-2	 80	Macular edema signal	1 approved 7 pipeline	81% of residues confidently predicted (no drug complex in this set)	Classes: Anti-VEGF intravitreal therapy
PPARA Peroxisome proliferator-activated receptor alpha	 77	Referable diabetic retinopathy	8 approved 2 pipeline	75% of residues confidently predicted (no drug complex in this set)	Classes: Fenofibrate (adjunct)
AKR1B1 Aldose reductase	 74	Referable diabetic retinopathy	2 approved 2 pipeline	drug-contact site pLDDT 98 (PDB 4JIR)	No guideline class yet
GLP1R Glucagon-like peptide-1 receptor	 73	Diabetes control; Blood pressure and vascular risk	7 approved 3 pipeline	drug-contact site pLDDT 86 (PDB 7K10)	Engaged by semaglutide. Classes: GLP-1 receptor agonist

Recruiting clinical trials

ClinicalTrials.gov, searched by finding near the clinic; no patient data is sent. The site confirms full eligibility.

Finding	Trial	Phase / sponsor	Pre-check
Referable diabetic retinopathy	A Study to Investigate the Safety, Tolerability, and Pharmacokinetics of RO7663498 Following Intravitreal Administration in Participants With Diabetic Retinopathy NCT07588100 Global Research Management, Glendale, California	Phase 1 · Hoffmann-La Roche	Age and sex fit; confirm full eligibility
Referable diabetic retinopathy	Implementing AI-based Glaucoma Screening Within Federally Qualified Health Centers NCT06629649 Department of Ophthalmology and Visual Sciences, University of Alabama at Birmingham, Birmingham, Alabama	NA · University of Alabama at Birmingham	Age and sex fit; confirm full eligibility
Referable diabetic retinopathy	Longitudinal Natural History Study of Retinal Function in Eyes of Patients With Diabetes NCT07270133 Southeast Retina Center, P.C., Augusta, Georgia	N/A · Jaeb Center for Health Research	Age and sex fit; confirm full eligibility
Macular edema signal	A Study in People With Diabetic Macular Edema to Test How Well Different Doses of BI 3812465 Are Tolerated NCT07553429 Associated Retina Consultants, Ltd., Phoenix, Arizona	Phase 1, Phase 2 · Boehringer Ingelheim	Age and sex fit; confirm full eligibility

Finding	Trial	Phase / sponsor	Pre-check
Macular edema signal	A Study to Evaluate the Safety, Tolerability, Pharmacodynamics, and Pharmacokinetics of RO7823653 in Participants With Diabetic Macular Edema (DME) NCT07425522 Associated Retina Consultants - Peoria - DocTrials - PPDS, Peoria, Arizona	Phase 1 · Genentech, Inc.	Age and sex fit; confirm full eligibility
Macular edema signal	A Clinical Trial of MK-8748 Compared to Aflibercept in Participants With Diabetic Macular Edema (MK-8748-004) NCT07705555 Global Research Management (Site 0066), Glendale, California	Phase 3 · Merck Sharp & Dohme LLC	Age and sex fit; confirm full eligibility
Blood pressure and vascular risk	Implementing AI-based Glaucoma Screening Within Federally Qualified Health Centers NCT06629649 Department of Ophthalmology and Visual Sciences, University of Alabama at Birmingham, Birmingham, Alabama	NA · University of Alabama at Birmingham	Age and sex fit; confirm full eligibility
Blood pressure and vascular risk	Improving the Collaborative Health of Adult Dyads NCT05370014 University of South Carolina, Columbia, South Carolina	NA · University of South Carolina	Age and sex fit; confirm full eligibility
Blood pressure and vascular risk	EASi-PROTKT™ - A Study to Test Vicadrostat (BI 690517) Taken Together With Empagliflozin in People With Type 2 Diabetes, High Blood Pressure, and Cardiovascular Disease NCT07064473 Pinnacle Research Group, LLC, Anniston, Alabama	Phase 3 · Boehringer Ingelheim	Age and sex fit; confirm full eligibility

CMS quality and billing

Measure / code	What it is	Fit for this encounter
CMS131 / MIPS Quality #117	Diabetes: Eye Exam	A screening read and signed by an eye care professional counts toward closing the gap. A signed specialist response in Insight Rx is the evidence to record.
HEDIS EED	Eye Exam for Patients With Diabetes	Closing EED gaps moves Star Ratings, which drive Medicare Advantage bonus payments to health plans.
CPT 92227	Retinal imaging, remote clinical staff review and report	Store-and-forward capture with review by clinical staff (no physician interpretation).
CPT 92228	Retinal imaging, remote physician or QHP interpretation and report	Insight Rx's workflow: photos captured at point of care, interpreted and signed by a physician.
CPT 92229	Retinal imaging, point-of-care automated analysis and report	Autonomous AI with an FDA-cleared device. Insight Rx is not FDA-cleared, so this does not apply today.

Methods, sources and limits

- Retinal models: DINOv2-L multi-task ensemble trained on mBRSET (portable camera, dilated); other cameras are a domain shift.
- Therapy classes and interactions are curated from ADA, AAO, KDIGO guidelines and FDA labels, cited per row; not a complete interaction checker.
- Targets: UniProt; structures: AlphaFold DB (CC-BY 4.0) and RCSB PDB; drugs: ChEMBL mechanisms; trials: ClinicalTrials.gov.
- Prepared for Dr. Alex Morgan. Patient identity in this workspace is synthetic.